Raymond Song CMU-educated Robotics/RL/ML Researcher

linkedin.com/in/rysong

raymond.song98@gmail.com | 8584650198 | California, US | US Citizen github.com/raymondsong00

TECHNICAL SKILLS

Python, C++, PyTorch, JAX, Reinforcement Learning (RL), Sim-to-Real Transfer, Deployed RL, Simulation (Isaac Sim, MuJoCo, Gazebo), Physical Hardware, Locomotion, Manipulation, Machine Learning, Distributed Training, Autonomous Vehicles, SLAM, Computer Vision, ROS/ROS2, CUDA, Docker, Linux, CMake, Git, Github, Writing Documentation

EXPERIENCE

Carnegie Mellon University | Robotics Researcher (Advised by Professor Jeff Schneider) **2024 - Present**

- Developed and published teacher-student **Reinforcement Learning** algorithm that extends PPO with teacher-guided exploration for sparse reward tasks, improves sample efficiency, and distills state-only policies into perception policies
- Trained simulation policy with novel RL algorithm, **PyTorch** and reward shaping to navigate difficult unstructured off-road driving tasks, outperforming RL (PPO, SAC), Imitation Learning (DAgger, GAIL, IQL) and MPC baselines
- Minimized the **Sim-to-Real** gap with domain randomization, high-fidelity simulation, and foundation models (DinoV2) and deployed an end-to-end policy zero-shot on a **physical robot**, achieving 100% obstacle avoidance in the field
- Designed large-scale policy training and hardware-in-the-loop (HIL) evaluation pipelines, custom reward functions, and tuned hyperparameters, tailoring decisions for real robot policy deployment and on-board compute constraints
- Integrated end-to-end low-level control and GNN RL policies seamlessly into C++ ROS autonomy stack via PyTorch compile, LibTorch, CUDA acceleration for real-time execution on National Robotics Engineering Center hardware
- Leveraged PyTorch profiler to isolate and eliminate bottlenecks to achieve 5x training speedup and save GPU compute
- Devised flexible parameter tuning interface with dynamic reconfiguration and ROS plugins for code reusability

AI Racing Tech (*$1.5M Autonomous Racecar*) | Various roles incl. Data Engineer, Deployment & Testing Lead **2023 - 2025**

- Created lightweight C++ ROS2 fault detection telemetry dashboard for autonomous Indy Car racing at 180MPH to ensure safety in a fast-paced and high-stakes environment, which reduced errors and resulted in 0 crashes over 2 years
- Performed safety-critical code review in large codebase, working closely with perception, planning, and control teams, and debugged issues from high-level software to low-level hardware for seamless transitions from code to deployment

TritonAI Autonomous Racing (*Autonomous Go-kart*) | Lead Robotics Engineer **2023 - 2024**

- Lead team of 5 engineers to build autonomous go-kart software via custom ROS2 Docker containers for Nvidia Jetson platforms, YOLOv8 detection for obstacles, GPS localization, and pure pursuit controller for speeds at 16MPH

PUBLICATIONS: *Several publications, only some are listed below*

- **R. Song***, Z. Wu*, V. Mundheda*, L. E. Navarro-Serment, C. Schoenborn, J. Schneider, "TADPO: Reinforcement Learning Goes Off-Road", ICRA 2026 **(Selected Oral Presentation, Top 11% of Papers)**
- T. Gupta*, A. Verma*, **R. Song**, D. Guttendorf, C. Igoe, L. E. Navarro-Serment, J. Schneider, "Efficient Active Search via Amortized Path-Integral Policies", ICRA 2026

OPEN SOURCE CONTRIBUTIONS (2020 - Present): *Several projects, only some are listed below*

TADPO: Teacher Action Distillation with Policy Optimization

- Teacher-student RL algorithm that integrates expert action distillation with on-policy learning to tackle poor exploration and sparse reward problems while maintaining training stability and avoiding behavior cloning's out of distribution errors

Chest X-Ray Expert Radiologist Report Generator

- Fine-tuned VLM (LLaVA) with LORA modules and context-embedded prompts to generate expert radiologist chest X-rays reports to help radiologists swiftly and accurately identify illnesses like pneumonia or pneumothorax

EDUCATION Carnegie Mellon University: Master of Science, Robotics | UC San Diego: BSc, Data Science **(Top 4%)**

Relevant Coursework: Robot Learning, Manipulation, Robot Safety, Deep Learning (CNN, Transformers, VLA), SLAM

PUBLIC TECH TALKS Master's Thesis: Real-World Off-Road Driving with RL (CMU), TADPO (ICRA 2026)

INTERESTS Reading novels by Dostoevsky | Getting lost in art museums | Steaks after strength training